

Monthly Research Trends: Developmental Cognitive Neuroscience

Report Date: 2026-05-17 **Period Covered:** 2026-05 (May 2026; recent publications included through 2026-05-17) **Verified**

Papers Included: 34

Note: Papers span January 2025 – May 2026. The rolling window captures recent publications added to the database through the report date.

1. Executive Summary

Top highlights this period:

- 1. Infants have rich visual categories in ventrotemporal cortex at 2 months** (*Nature Neuroscience*): O'Doherty, Cusack et al. used awake fMRI in >100 two-month-old infants to show that categorical representations for 12 object classes are already established in high-level visual cortex, closely mirroring adult patterns. DOI: [10.1038/s41593-025-02187-8](https://doi.org/10.1038/s41593-025-02187-8)
- 2. Income insufficiency impacts early EEG brain networks in infants** (*PNAS*): Chung, Wilkinson, Nelson et al. (Boston Children's Hospital / Harvard) used network-based EEG analysis to show that income-related adversity alters resting-state oscillatory network topology from 4–12 months. DOI: [10.1073/pnas.2513598123](https://doi.org/10.1073/pnas.2513598123)
- 3. Caregiver-infant interactions shape functional connectivity at 1 month** (*bioRxiv*, *Lloyd-Fox / M. H. Johnson*): Carnevali et al. (Birkbeck) combined fNIRS brain imaging with coded dyadic interactions at one month and found

affectionate touch and caregiver engagement predicted stronger social brain network connectivity. DOI: [10.64898/2026.05.13.724776](https://doi.org/10.64898/2026.05.13.724776)

4. **Explicitly nonlinear fMRI reveals hidden infant developmental trajectories** (*bioRxiv*): Kinsey, Iraj et al. present the first whole-brain nonlinear ICA fMRI analysis in infants, uncovering sensorimotor, default mode, executive, and language networks invisible to standard linear methods. DOI: [10.64898/2026.04.07.716703](https://doi.org/10.64898/2026.04.07.716703)
5. **Aperiodic EEG changes predict autism diagnosis and language at 18 months** (*Autism Research*): Wilkinson, Chung, Nelson et al. (BCH/Harvard) show that greater increases in aperiodic offset and slope from 3–12 months predict both later autism diagnosis and worse language outcomes in high-risk infants. DOI: [10.1002/aur.70063](https://doi.org/10.1002/aur.70063)

2. Quick Reference Table

#	Paper	Institution	Method	Age	Preprint?	★
1	Carnevali et al. — <i>bioRxiv</i>	Birkbeck, Univ. of London	fNIRS	1 month	bioRxiv	★
2	Khandelwal et al. — <i>bioRxiv</i>	MGH / Harvard Med.	fMRI (postmortem)	Neonatal	bioRxiv	
3	Zou et al. — <i>bioRxiv</i>	Trinity College Dublin	fMRI (RS)	Neonates → 18 mo	bioRxiv	
4	Kinsey et al. — <i>bioRxiv</i>	Georgia State / TReNDS	fMRI (nonlinear)	0–12 months	bioRxiv	
5	Vanneau et al. — <i>bioRxiv</i>	Albert Einstein / Univ. Rochester	EEG (HD)	8–14 years	bioRxiv	
6	Yan et al. — <i>bioRxiv</i>	McGill University	Mixed	9–15 years	bioRxiv	
7	Zou et al. — <i>bioRxiv</i>	Trinity College Dublin	fMRI (RS)	Neonates → 18 mo	bioRxiv	
8	[Authors] et al. — <i>bioRxiv</i>	Multiple	fMRI (RS)	10 mo–12 years	bioRxiv	
9	Ellwood-Lowe et al. — <i>bioRxiv</i>	Univ. of Pennsylvania	fMRI (Hurst)	10 mo–15 years	bioRxiv	
10	[Authors] et al. — <i>bioRxiv</i>	dHCP consortium	fMRI (RS)	Neonates	bioRxiv	

#	Paper	Institution	Method	Age	Preprint?	★
11	[Authors] et al. — <i>bioRxiv</i>	[Unverified]	fNIRS	0–12 months	bioRxiv	
12	Green et al. — <i>Dev. Cogn. Neurosci.</i>	CHOP / Univ. Pennsylvania	MEG	2–68 months	No	
13	Leroy et al. — <i>PNAS</i>	CEA, France	EEG	4 months	No	
14	Dickinson et al. — <i>bioRxiv</i>	IBIS Network	EEG	0–12 months	bioRxiv	
15	[Authors] et al. — <i>bioRxiv</i>	[Unverified]	fMRI (task)	19–36 months	bioRxiv	
16	Beaugrand et al. — <i>npj Bio Timing Sleep</i>	Univ. of Surrey	EEG (sleep)	3–6 months	No	
17	Hardiansyah et al. — <i>Sci. Reports</i>	Karolinska / Uppsala	EEG	5 months	No	
18	[Authors] et al. — <i>bioRxiv</i>	[Unverified]	EEG	0–12 months	bioRxiv	
19	Sa de Almeida et al. — <i>Nature Comms.</i>	[Unverified]	fMRI (RS)	Neonates/ preterm	No	
20	Epihova et al. — <i>J. Neuroscience</i>	Univ. of Cambridge	fMRI	Fetal	No	★
21	Kuriyakose et al. — <i>Front. Comp. Neurosci.</i>	Vellore Institute of Tech.	EEG	Children/ adults	No	
22	Brandes-Aitken et al. — <i>Dev. Cogn. Neurosci.</i>	New York University	EEG	3 months	No	
23	O'Doherty et al. — <i>Nature Neurosci.</i> ★	Trinity College Dublin	fMRI (task)	2 months	No	
24	Gilbreath et al. — <i>Psychophysiology</i>	Univ. of Arkansas / UAMS	EEG	2–12 months	No	
25	[Authors] et al. — <i>Comms. Biology</i>	dHCP consortium	fMRI (RS)	Neonates	No	
26	[Authors] et al. — <i>Nature Comms.</i>	[Unverified]	fMRI	Birth–childhood	No	
27	Greenhalgh et al. — <i>Neurophotonics</i>	Univ. of Cambridge	fNIRS	1 month	No	★
28	Li et al. — <i>Comms. Biology</i>		fMRI (RS)	0–6 months	No	

#	Paper	Institution	Method	Age	Preprint?	★
		Georgia State / TReNDS				
29	Marino et al. — <i>Comms. Psychology</i>	[Unverified]	fNIRS	Neonates	No	
30	Chung et al. — <i>PNAS</i> ★	BCH / Harvard Med.	EEG	4–12 months	No	★
31	Chen et al. — <i>Molecular Autism</i>	BCH / Harvard Med.	EEG	2–6 years	No	★
32	[Authors] et al. — <i>bioRxiv</i>	[Unverified]	EEG + fMRI	Neonates	bioRxiv	
33	Wang et al. — <i>Dev. Cogn. Neurosci.</i>	Univ. of Chicago	EEG	8–12 months	No	★
34	Wilkinson et al. — <i>Autism Research</i> ★	BCH / Harvard Med.	EEG	0–12 months	No	★

★ = *priority author paper*

3. Research Focus Guide

If you work on...	Read these papers
EEG: aperiodic slope / FOOOF	12, 24, 31, 34
EEG: oscillatory rhythms & power	5, 13, 16, 18, 22, 33
EEG: early adversity & environment	22, 30
EEG: autism biomarkers	5, 14, 17, 21, 31, 34
fMRI: neonatal & infant RS connectivity	3, 4, 7, 10, 19, 25, 26, 28, 32
fMRI: awake infant / task-based	23, 15
fMRI: developmental trajectories	8, 9, 20
fNIRS: early development	1, 11, 27, 29
Language development	9, 15, 29, 33
Autism early markers & trajectories	5, 14, 17, 21, 31, 34
Early adversity / SES / caregiving	1, 6, 22, 30
Normative developmental trajectories	9, 12, 16, 18, 24, 26, 28
Multivariate & ML / nonlinear methods	4, 6, 10, 11, 21, 25
Cross-cultural / global neuroscience	27
Fetal & prenatal brain	20
Infant brain atlas / methods	2

4. Papers by Topic

4.1 EEG: Early Development, Adversity & Environment

Paper 16 Title: Tracing infant sleep neurophysiology longitudinally from 3 to 6 months: EEG insights into brain development

 **EEG (high-density, sleep) · 3–6 months · Sleep neurophysiology · SWA · Thalamocortical maturation**
 **Read if you study:** infant sleep EEG, cortical maturation markers, or normative oscillatory development in the first year  **In one sentence:** Slow-wave activity increased occipitally, sigma power shifted from central to frontal, and theta increased globally from 3 to 6 months — providing frequency-specific markers of thalamocortical development.

Authors: Matthieu Beaugrand, Valeria Jaramillo, Christophe Mühlematter, Sarah F. Schoch, Vivien Reicher, Andjela Markovic, Salome Kurth **Journal:** *npj Biological Timing and Sleep* | **Date:** 2026-03-01 | **Preprint:** No **DOI:** <https://doi.org/10.1038/s44323-026-00071-7> **First Author Institution:** University of Surrey, Surrey Sleep Research Centre **Age Range:** 3–6 months | **Method:** EEG (high-density, NREM sleep) | **Statistical Approach:** EEG power spectral analysis; topographic mapping; longitudinal mixed-effects models **Key Findings:** Developmental shifts in sleep EEG topography from 3 to 6 months provide normative benchmarks for sleep EEG maturation across frequency bands. SWA, theta, and sigma each showed distinct topographic trajectories consistent with progressive cortical and thalamocortical maturation. **Key Figure:** Topographic maps of EEG power across sleep stages at 3 and 6 months.

Paper 24 Title: Effects of Early Infant Nutrition on Aperiodic Exponent From 2 to 12 Months Using EEG Analysis

 **EEG (resting-state) · 2–12 months · Aperiodic slope · Nutrition · E/I balance** 📌 **Read if you study:**

EEG biomarkers of brain development, E/I balance maturation, or effects of early nutrition on the brain 💡

In one sentence: Breastfed infants had a higher aperiodic exponent than formula-fed infants at 3 months only — nutrition-group differences did not persist to any other timepoint from 2 to 12 months.

Authors: Dylan Gilbreath, Adam Andrews, Darcy Hagood, Aline Andres, Linda J Larson-Prior **Journal:** *Psychophysiology* | **Date:** 2026-02-01 | **Preprint:** No **DOI:** <https://doi.org/10.1111/psyp.70247> **First Author Institution:** University of Arkansas for Medical Sciences **Age Range:** 2–12 months | **Method:** EEG | **Statistical Approach:** Power spectral density (Specparam/FOOOF); mixed-effects models by feeding group **Key Findings:** Aperiodic exponent decreased with age across all infants. Breastfeeding showed a transient advantage at 3 months only. One of the first infant EEG studies applying Specparam to longitudinal nutrition data. **Key Figure:** Longitudinal aperiodic exponent trajectories by feeding group across 2, 6, and 12-month timepoints.

Paper 22 Title: Unpredictable caregiving is associated with disrupted neurophysiological measures of attention and autonomic function in three-month-old infants

 **EEG (frontal theta) · 3 months · Early adversity · Caregiving · Attention · HRV** 📌 **Read if you study:**

early adversity effects on infant neurodevelopment, caregiving quality, or neural markers of attention and autonomic regulation 💡 **In one sentence:** Infants exposed to more unpredictable caregiving at 3 months show

disrupted frontal theta power and heart rate variability — early neural signs of how caregiving quality shapes attentional and autonomic systems.

Authors: Annie Brandes-Aitken, Maya Metser, Stephen H. Braren, Sarah C. Vogel, Natalie H. Brito **Journal:** *Developmental Cognitive Neuroscience* | **Date:** 2026-02-04 | **Preprint:** No **DOI:** <https://doi.org/10.1016/j.dcn.2026.101688> **First Author Institution:** New York University, Department of Applied Psychology **Age Range:** 3 months | **Method:** EEG | **Statistical Approach:** Frontal theta power analysis; heart rate variability (HRV) analysis; regression models **Key Findings:** Unpredictable caregiving environments linked to disrupted EEG-based attention markers and reduced HRV at 3 months, highlighting early neural consequences of caregiving adversity. **Key Figure:** Correlation between caregiving unpredictability scores and infant frontal theta power/HRV metrics.

Paper 30 ★ *Top Pick* — *Priority Author* **Title:** Income insufficiency impacts early brain development in infants facing increased psychosocial adversity: A network-based approach

🔬 **EEG (resting-state, network-based) · 4–12 months**
· **Early adversity · SES · Neural networks · Graph metrics** 📌 **Read if you study:** socioeconomic effects on infant brain development, early adversity, or EEG network-based analysis 💡 **In one sentence:** Income-related adversity is associated with altered resting-state oscillatory network topology from 4–12 months, revealing network-level neural mechanisms through which socioeconomic factors shape the developing brain.

Authors: Haerin Chung, Carol L. Wilkinson, A. Liu, A. Job Said, B. Francis, G. Cañaveral, K. Conroy, Helen Tager-Flusberg, Charles A. Nelson **Journal:** *Proceedings of the National Academy of Sciences* | **Date:** 2026-01-13 | **Preprint:** No **DOI:** <https://doi.org/10.1073/pnas.2513598123> **First Author Institution:** Boston

Children's Hospital, Harvard Medical School **Age Range:** 4–12 months | **Method:** EEG | **Statistical Approach:** Partial correlation network analysis; longitudinal EEG power analysis; graph-theoretic network metrics **Key Findings:** Network-based EEG analysis reveals that income insufficiency and psychosocial adversity alter the topology of resting-state oscillatory networks from 4 to 12 months, providing mechanistic insight into how early socioeconomic factors shape infant brain development. **Key Figure:** Network connectivity matrices comparing income-sufficient versus income-insufficient groups across development.

Paper 33 ★ *Priority Author* **Title:** Language exposure predicts infants' neural processing of others' actions based on language group

🔬 **EEG (mu desynchronization, frontal theta, alpha asymmetry) · 8–12 months · Language exposure · Action observation · Social cognition** 📌 **Read if you study:** language effects on social brain development, motor resonance, or early social cognition 💡 **In one sentence:** Language group moderates infant neural responses (mu desynchronization and frontal theta) during action observation at 8–12 months, showing early linguistic experience shapes the neural systems underlying action understanding.

Authors: Yiyi Wang, Marc Colomer, Hyesung Grace Hwang, Enda Tan, Nathan A Fox **Journal:** *Developmental Cognitive Neuroscience* | **Date:** 2026-01-01 | **Preprint:** No **DOI:** <https://doi.org/10.1016/j.dcn.2025.101645> **First Author Institution:** University of Chicago **Age Range:** 8–12 months | **Method:** EEG | **Statistical Approach:** Mu desynchronization analysis; frontal theta power; frontal alpha asymmetry; mixed-effects models **Key Findings:** Early linguistic experience shapes neural systems for social cognition — language group predicted infants' motor

resonance responses during observation of goal-directed actions.

Key Figure: EEG mu desynchronization topographic maps by language group during action observation.

Paper 13 Title: Dynamics of the attentional blink in preverbal infants

 **EEG (steady-state, frequency-tagging) · 4 months · Visual attention · Attentional blink · Temporal dynamics** 📌 **Read if you study:** infant attention, temporal dynamics of visual processing, or EEG frequency-tagging methods in infants 💡 **In one sentence:** Four-month-old infants show attentional blink dynamics analogous to those in adults — revealing that temporal constraints on visual attention are present and measurable before the first birthday.

Authors: F. Leroy, S. Naik, R. Gulbinaite, M. Palu, D. Battaglia, G. Dehaene-Lambertz **Journal:** *Proceedings of the National Academy of Sciences* | **Date:** 2026-03-10 | **Preprint:** No **DOI:** <https://doi.org/10.1073/pnas.2526752123> **First Author Institution:** CEA (Commissariat à l'énergie atomique), France **Age Range:** 4 months | **Method:** EEG | **Statistical Approach:** Saccade analysis; pupillometry; steady-state EEG frequency-tagging; temporal dynamics modeling **Key Findings:** Preverbal 4-month-old infants exhibit attentional blink dynamics — a temporal limitation in visual attention — consistent with adult patterns, demonstrating early-emerging constraints on temporal attention. **Key Figure:** Attentional blink curves (detection accuracy vs. inter-target lag) in 4-month-old infants compared to adult benchmarks.

Paper 18 Title: From Variability to Synchrony: Non-linear Development of Auditory Neural Responses During the First Year of Life

 **EEG (ITPC, high-density) · 2 weeks–12 months · Auditory processing · Phase synchrony · Non-linear trajectory**  **Read if you study:** auditory cortex development, longitudinal EEG in infancy, or non-linear developmental trajectories  **In one sentence:** Auditory phase synchrony increases from 2 weeks to 6 months then unexpectedly decreases from 6 to 12 months — revealing a non-linear developmental transition toward more differentiated brain activity in the first year.

Authors: [Unverified — full author list at DOI] **Journal:** *bioRxiv* | **Date:** 2026-02-20 | **Preprint:** Yes (bioRxiv) **DOI:** <https://doi.org/10.64898/2026.02.20.706969> **Age Range:** 0–12 months (2 weeks, 6 months, 12 months; N=69) | **Method:** EEG | **Statistical Approach:** Inter-trial phase coherence (ITPC); longitudinal mixed models **Key Findings:** Neural responses to pure tones became more pronounced and less variable with age. Phase synchronization increased then decreased, suggesting a non-linear developmental shift from global synchrony to differentiated responses across the first year.

Paper 12 Title: Resting-state periodic and aperiodic brain oscillations from birth to preschool years: Aperiodic maturity predicts developmental course

 **MEG (resting-state) · 2–68 months · Aperiodic slope · Normative trajectories · Adaptive behavior**  **Read if you study:** normative oscillatory trajectories from birth to preschool, aperiodic exponent as a developmental biomarker, or MEG in young children  **In one sentence:** More mature (steeper) aperiodic values at earlier ages predicted better adaptive behaviors and daily living skills — establishing aperiodic MEG as a clinically meaningful birth-to-preschool biomarker.

Authors: Heather L Green, James Christopher Edgar, Kylie Mol, Marybeth McNamee, Laura Prosser, Mina Kim, Emily S Kushner, Gregory A Miller, Yuhan Chen **Journal:** *Developmental Cognitive Neuroscience* | **Date:** 2026-03-14 | **Preprint:** No **DOI:** <https://doi.org/10.1016/j.dcn.2026.101709> **First Author Institution:** Children's Hospital of Philadelphia, Department of Radiology **Age Range:** 2–68 months | **Method:** MEG | **Statistical Approach:** Specparam (FOOOF); power spectrum analysis; nonlinear age modeling; mixed-effects models **Key Findings:** Aperiodic maturity — the degree to which the exponent approaches adult-like values — predicts subsequent developmental course across birth through the preschool years. Separate modeling of aperiodic and periodic components reveals distinct developmental trajectories per component. **Key Figure:** Aperiodic exponent developmental trajectories from birth to preschool with age-nonlinear curve fits.

4.2 EEG: Autism & Clinical Populations

Paper 34 ★ *Top Pick* — *Priority Author* **Title:** Changes in Early Aperiodic EEG Activity Are Linked to Autism Diagnosis and Language Development in Infants With Family History of Autism

🔬 **EEG (resting-state, aperiodic) · 0–12 months · Autism · Language · Aperiodic trajectory · High-risk infants** 📌 **Read if you study:** EEG biomarkers for autism prediction, early language development in high-risk populations, or longitudinal aperiodic EEG 💡 **In one sentence:** Greater increases in aperiodic offset and slope from 3 to 12 months predicted both later autism diagnosis and worse 18-month language outcomes in infants with elevated familial likelihood for autism.

Authors: Carol L. Wilkinson, Haerin Chung, Amy Dave, Helen Tager-Flusberg, Charles A. Nelson **Journal:** *Autism Research* | **Date:** 2025-07-15 | **Preprint:** No **DOI:** <https://doi.org/10.1002/aur.70063> **First Author Institution:** Boston Children's Hospital, Harvard Medical School **Age Range:** 0–12 months (3–12 months

longitudinal EEG; language at 18 months) | **Method:** EEG | **Statistical Approach:** Longitudinal EEG aperiodic change (3–12 months); regression on ASD diagnosis and 18-month language outcomes **Key Findings:** Infants with a later autism diagnosis showed significantly increased aperiodic activity change vs. high-risk non-diagnosed infants. Dynamic changes in aperiodic EEG serve as early indicators of both autism diagnosis and language outcome, even when controlling for autism diagnosis.

Paper 31 ★ *Priority Author* **Title:** Resting state aperiodic and periodic EEG activity in preschool-aged autistic children: differences from neurotypical peers and links to language skills

🔬 **EEG (resting-state, aperiodic + periodic) · 2–6 years · Autism · Aperiodic slope · Language · Preschool** 📌 **Read if you study:** EEG spectral features in autistic children, language development in autism, or aperiodic/periodic EEG in preschoolers 💡 **In one sentence:** Both aperiodic and periodic EEG features differed between 64 autistic and 64 neurotypical preschoolers aged 2–6, and were significantly associated with concurrent language ability.

Authors: Yanru Chen, Meagan Tsou, Charles A. Nelson, Helen Tager-Flusberg, Carol L. Wilkinson **Journal:** *Molecular Autism* | **Date:** 2026-01-10 | **Preprint:** No **DOI:** <https://doi.org/10.1186/s13229-025-00700-1> **First Author Institution:** Boston Children's Hospital, Harvard Medical School **Age Range:** 2–6 years | **Method:** EEG | **Statistical Approach:** Spectral parameterization (specparam/FOOOF); group comparison (64 ASD, 64 neurotypical); regression on language skills **Key Findings:** Aperiodic and periodic EEG features show group differences between autistic and neurotypical preschoolers and link to language ability — pointing to spectral EEG as a neural mechanism marker for language development in autism.

Paper 14 Title: Visual Cortical Response Variability in Infants at High Familial Likelihood for Autism

 **EEG (VEP, high-density) · 0–12 months · Autism risk · Visual cortex · P1 variability · IBIS Network** 

Read if you study: early EEG biomarkers of autism, infant visual processing, or trial-by-trial neural variability in high-risk populations  **In one sentence:** Greater trial-to-trial variability in P1 latency at 6 and 12 months counterintuitively predicted higher cognitive and language scores at 24 months — suggesting flexible sensory processing confers developmental advantage in infants at elevated autism risk.

Authors: Abigail Dickinson, Madison Booth, Scott Huberty, Declan Ryan, Alana Campbell, Jessica B. Girault, Neely Miller, Bonnie Lau, John Zempel, Sara Jane Webb, Jed Elison, Adrian K.C. Lee, Annette Estes, Stephen R. Dager, Heather Hazlett, Jason Wolff, Robert Schultz, Natasha Marrus, Alan Evans, Joseph Piven, John R. Pruett Jr., Shafali Jeste **Journal:** *bioRxiv* | **Date:** 2026-03-09 | **Preprint:** Yes (bioRxiv) **DOI:** <https://doi.org/10.64898/2026.03.05.709374> **First Author Institution:** IBIS Network (multiple institutions) **Age Range:** 0–12 months (VEPs at 6 & 12 months; Bayley at 24 months) | **Method:** EEG | **Statistical Approach:** VEP P1 latency variability; group comparison; regression on Bayley-III at 24 months **Key Findings:** P1 latency variability emerged as the strongest neural correlate of later developmental outcomes, suggesting adaptive sensory circuit flexibility facilitates experience-dependent tuning in high-risk infants.

Paper 17 Title: Functional connectivity in infants' visual cortex and its links to motion processing and autism

 **EEG (gamma-band connectivity) · 5 months · Autism risk · Visual cortex · Gamma connectivity · EASE cohort**  **Read if you study:** early EEG biomarkers

of autism, visual cortex development, or gamma connectivity in infancy 💡 **In one sentence:** Gamma EEG connectivity between midline and lateral visual cortex at 5 months predicted both later autism symptoms and global motion cortical laterality — identifying a pre-symptomatic autism biomarker.

Authors: Irzam Hardiansyah, Giorgia Bussu, Sven Bölte, Emily J. H. Jones, Terje Falck-Ytter **Journal:** *Scientific Reports* | **Date:** 2026-02-28 | **Preprint:** No **DOI:** <https://doi.org/10.1038/s41598-026-42048-3> **First Author Institution:** Karolinska Institutet / Uppsala University **Age Range:** 5 months | **Method:** EEG | **Statistical Approach:** EEG functional connectivity (theta, alpha, gamma); EASE cohort (N=59, 39 elevated familial likelihood) **Key Findings:** Altered early visual cortical gamma connectivity may link visual motion processing atypicalities to later autism symptomatology, making it a candidate EEG biomarker measurable before behavioral symptoms emerge.

Paper 5 Title: Instability of Alpha Oscillatory States in Autism and Familial Liability: Evidence from Burst-Resolved High-Density EEG

🔬 **EEG (HD, burst-resolved alpha) · 8–14 years · Autism · Alpha stability · Familial liability · Burst dynamics** 📌 **Read if you study:** EEG oscillatory mechanisms in autism, alpha rhythm dynamics, or familial risk gradients in ASD 💡 **In one sentence:** Reduced alpha power in autism reflects reduced temporal stability of alpha bursts (not lower per-burst amplitude), and siblings show intermediate instability — revealing a familial liability gradient in alpha burst dynamics.

Authors: Theo Vanneau, Chloe Brittenham, Megan Darrell, Michael Quiquempoix, John J. Foxe, Sophie Molholm **Journal:** *bioRxiv* | **Date:** 2026-04-07 | **Preprint:** Yes (bioRxiv) **DOI:** <https://doi.org/10.64898/2026.04.03.716324> **First Author Institution:** Albert Einstein College of Medicine / University of

Rochester **Age Range:** 8–14 years (*outside 0–36 month focus; included for autism oscillatory & familial liability relevance*)

Method: EEG (HD) | **Statistical Approach:** Burst-resolved alpha analysis; time-frequency decomposition; group comparison (n=39 non-autistic, n=52 autistic, n=26 siblings) **Key Findings:** The mechanistic distinction — stability vs. amplitude — has important implications for EEG biomarker development and understanding familial risk endophenotypes in autism.

Paper 21 Title: Explainable AI uncovers novel EEG microstate candidate neurophysiological markers for autism spectrum disorder

 **EEG (microstates, ML) · Children/adults · Autism · EEG microstates · XGBoost · Explainable AI** 📌 **Read if you study:** EEG-based autism classification, machine learning approaches to neurophysiological biomarkers, or interpretable AI in neuroscience 💡 **In one sentence:** An XGBoost classifier using multi-domain EEG microstate features (temporal, spectral, complexity, graph-theoretic) achieved 80.87% ASD classification accuracy, with SHAP analysis identifying fractional occupancy of microstate 3 and delta-band power as top discriminators.

Authors: Delna Kuriyakose, Gowsalya M. **Journal:** *Frontiers in Computational Neuroscience* | **Date:** 2026-02-04 | **Preprint:** No **DOI:** <https://doi.org/10.3389/fncom.2026.1763727> **First Author Institution:** Vellore Institute of Technology, India **Age Range:** Children/adults | **Method:** EEG | **Statistical Approach:** EEG microstates; XGBoost (80.87% accuracy); SHAP explainability; multi-domain features **Key Findings:** Interpretable neurophysiological biomarkers for ASD identified via SHAP analysis, providing a novel multi-domain analytical pipeline with clinical potential for ASD detection.

4.3 fMRI: Neonatal, Infant & Structural

Paper 23 ★ *Top Pick* **Title:** Infants have rich visual categories in ventrotemporal cortex at 2 months of age

 **fMRI (awake, task-based) · 2 months · Visual categorization · RSA · Ventrotemporal cortex** 📌 **Read if you study:** infant visual cortex development, representational structure in early life, or awake infant fMRI methodology 💡 **In one sentence:** Categorical representations for 12 visual object classes — closely matching adult patterns — are already present in infant ventrotemporal cortex at just 2 months of age.

Authors: Cliona O'Doherty, Áine T. Dineen, Anna Truzzi, Graham King, Lorijn Zaadnoordijk, Keelin Harrison, Enna-Louise D'Arcy, Jessica White, Chiara Caldinelli, Tamrin Holloway, Anna Kravchenko, Jörn Diedrichsen, Ailbhe Tarrant, Angela T. Byrne, Adrienne Foran, Eleanor J. Molloy, Rhodri Cusack **Journal:** *Nature Neuroscience* | **Date:** 2026-02-02 | **Preprint:** No **DOI:** <https://doi.org/10.1038/s41593-025-02187-8> **First Author Institution:** Trinity College Dublin **Age Range:** 2 months (n>100; follow-up at 9 months) | **Method:** fMRI (awake, task-based) | **Statistical Approach:** Representational similarity analysis (RSA); deep neural network comparison; multivariate pattern analysis **Key Findings:** Category representations in ventrotemporal cortex at 2 months were similar to adult representations including animate/inanimate distinctions. A deep neural network model aligned with infants' representational geometry, indicating early category templates can emerge from visual input statistics.

Paper 20 ★ *Priority Author* **Title:** The Fronto-Temporal Cortex Has Increased Subcortical Connectivity In Utero and Plasticity in Adulthood

 **fMRI (fetal/perinatal) · Fetal (24 weeks GA) · Fronto-temporal · Subcortical connectivity · Plasticity · Myelination** 📌 **Read if you study:** prenatal brain development, fronto-temporal cortex evolution, or the

developmental basis of cortical plasticity 💡 **In one sentence:** At 24 gestational weeks, fronto-temporal regions show disproportionate subcortical connectivity and lower adult myelination — suggesting an evolved extended developmental timeline to maximize flexible neural responses.

Authors: Gabriela Epihova, Dimitar Z. Epihov, Danyal Akarca, Duncan E. Astle **Journal:** *Journal of Neuroscience* | **Date:** 2026-02-12 | **Preprint:** No **DOI:** <https://doi.org/10.1523/JNEUROSCI.0832-25.2025> **First Author Institution:** MRC Cognition and Brain Sciences Unit, University of Cambridge **Age Range:** Fetal/prenatal | **Method:** fMRI | **Statistical Approach:** Fetal MRI connectivity analysis; gene expression annotation; perineuronal net markers; myelination quantification **Key Findings:** Fronto-temporal cortex shows heightened progenitor cell markers and lower perineuronal net gene expression in adulthood, consistent with protracted plasticity as an evolutionarily derived feature supporting flexible postnatal learning.

Paper 2 Title: Cortical reconstruction and anatomical parcellation of high-resolution multi-modal postmortem ex vivo MRI of the human infant brain

🔬 **fMRI (postmortem, ex vivo) · Neonatal · Brain atlas · Deep learning segmentation · Cortical reconstruction** 📌 **Read if you study:** infant brain MRI methodology, cortical atlases, or postmortem neuroanatomy tools 💡 **In one sentence:** PostSynth — a postmortem-specific synthetic data engine — enables automated cortical surface reconstruction in postmortem infant brains by modeling fixation-driven imaging characteristics (altered gray-white contrast, laminar heterogeneity, bias fields).

Authors: Pulkit Khandelwal, Sala Young, Nathan Xi Ngo, Paul A. Yushkevich, Andre van der Kouwe, Robin L. Haynes, Hannah C. Kinney, Lilla Zöllei **Journal:** *bioRxiv* | **Date:** 2026-05-09 | **Preprint:** Yes (bioRxiv) **DOI:** <https://doi.org/10.64898/2026.05.07.722301> **First Author Institution:** Massachusetts General Hospital / Harvard Medical School **Age Range:** Neonatal/postmortem | **Method:** fMRI (ex vivo) | **Statistical Approach:** Deep learning volumetric segmentation; PostSynth synthetic data generation; cortical surface reconstruction **Key Findings:** The framework enables anatomical segmentation and cortical surface reconstruction in postmortem infant brains — creating a publicly available tool for studying early cortical architecture.

Paper 19 Title: Regional BOLD variability reflects microstructural maturation and neuronal ensheathment in the preterm infant cortex

 **fMRI (resting-state BOLD variability) · Neonates/preterm · BOLD variability · Microstructural maturation · Ensheathment** 📌 **Read if you study:** preterm brain development, fMRI biomarkers of cortical maturation, or BOLD signal variability 💡 **In one sentence:** BOLD variability increases in preterm infants alongside concurrent decreases in cortical diffusivity and upregulation of gliogenesis/ensheathment genes — linking fMRI signal variability to ongoing microstructural maturation.

Authors: Sa de Almeida J., Boehringer A., Loukas S., and co-authors **Journal:** *Nature Communications* | **Date:** 2026-02-15 | **Preprint:** No **DOI:** <https://doi.org/10.1038/s41467-026-71415-x> **Age Range:** 0–1 month neonatal/preterm (N=54 preterm + 24 full-term) | **Method:** fMRI (RS) | **Statistical Approach:** Longitudinal BOLD variability analysis; multi-shell diffusion MRI; gene expression analysis **Key Findings:** By term age, preterm infants showed immature cortical maturation patterns compared

with full-term newborns. BOLD variability links to both microstructural and molecular indices of ongoing cortical development.

Paper 25 Title: Aperiodic parameters of the fMRI power spectrum associate with preterm birth and neonatal age

 **fMRI (resting-state, aperiodic BOLD) · Neonates · Aperiodic fMRI · Brain maturity · Preterm · dHCP** 
Read if you study: neonatal brain maturation biomarkers, BOLD signal aperiodic structure, or preterm brain development  **In one sentence:** Aperiodic parameters of the BOLD power spectrum — exponent and offset from pre-/postcentral gyrus — associate with preterm birth and neonatal age, explaining $R^2=0.20-0.41$ of age variance.

Authors: [Unverified — dHCP consortium] **Journal:** *Communications Biology* | **Date:** 2026-01-23 | **Preprint:** No **DOI:** <https://doi.org/10.1038/s42003-025-09488-5> **Age Range:** Neonates | **Method:** fMRI (RS) | **Statistical Approach:** Aperiodic BOLD signal parameterization (0.01–0.15 Hz); 90 cortical/subcortical ROIs; machine learning regression; dHCP dataset **Key Findings:** Demonstrates feasibility of studying functional brain maturation via fMRI-based aperiodic activity. Pre- and postcentral gyri BOLD aperiodic parameters are sensitive biomarkers of prematurity and age.

Paper 26 Title: Hierarchical maturation of structural brain connectomes from birth to childhood

 **fMRI (structural connectome) · Birth–childhood · Structural connectomes · Network maturation · Hub consolidation**  **Read if you study:** structural brain development, connectome maturation, or prefrontal network development from infancy  **In one sentence:** Two cardinal topological transformations characterize structural brain maturation from birth to childhood: global

nonlinear increases in network efficiency, and regional hub consolidation with prolonged modular reconfiguration in prefrontal and insular cortices.

Authors: [Unverified — full author list at DOI] **Journal:** *Nature Communications* | **Date:** 2026-01-22 | **Preprint:** No **DOI:** <https://doi.org/10.1038/s41467-026-68704-w> **Age Range:** Birth–childhood | **Method:** fMRI | **Statistical Approach:** Connectome mapping; machine learning; neurobiological annotation; network efficiency metrics **Key Findings:** Association cortices show distinct sensorimotor-association developmental trajectories; early developmental trajectories of association cortices predict late childhood network architecture via microstructural maturation of connected white matter tracts.

Paper 28 Title: Deciphering multiway multiscale brain network connectivity from birth to 6 months

 **fMRI (resting-state, higher-order connectivity) · 0–6 months · Triadic interactions · Network architecture · Triple network** 📌 **Read if you study:** neonatal brain network organization, higher-order functional connectivity, or the origins of canonical adult networks 💡 **In one sentence:** Triadic functional interactions among default mode, sensorimotor, visual, limbic, language, salience, and executive networks — mirroring the adult triple network model — are already established in the first six postnatal months.

Authors: Qiang Li, Zening Fu, Hasse Walum, Masoud Seraji, Prerana Bajracharya, Vince D. Calhoun, Sarah Shultz, Armin Iraj **Journal:** *Communications Biology* | **Date:** 2026-01-16 | **Preprint:** No **DOI:** <https://doi.org/10.1038/s42003-026-09549-3> **First Author Institution:** Georgia State University / Emory University / TReNDS Center **Age Range:** 0–6 months (N=71 infants, 126 scans) | **Method:** fMRI (RS) | **Statistical Approach:** High-order triadic interactions; beyond-pairwise functional

connectivity; functional network connectivity **Key Findings:** High-order triadic interactions provided more reliable biomarkers of developmental trajectories than pairwise measures, suggesting hierarchical, multiway brain functional architecture is a fundamental feature from the earliest postnatal weeks.

Paper 3 Title: Neonatal Resting-State Functional Connectivity Predicts Socioemotional and Behavioral Outcomes at 18 Months

 **fMRI (resting-state, CPM) · Neonates → 18 months**
· **Connectome-based prediction · Socioemotional outcomes · dHCP**  **Read if you study:** neonatal fMRI, early prediction of behavioral outcomes, or connectome-based predictive modeling  **In one sentence:** Neonatal whole-brain resting-state connectivity significantly predicted socioemotional temperament at 18 months (externalizing, surgency, negative affect, effortful control), with distinct predictive architectures for term vs. preterm infants.

Authors: Mi Zou, Arun L. W. Bokde **Journal:** *bioRxiv* | **Date:** 2026-04-21 | **Preprint:** Yes (bioRxiv) **DOI:** <https://doi.org/10.64898/2026.04.21.719787> **First Author Institution:** Trinity College Dublin **Age Range:** Neonates → 18 months (N=397: 277 term, 120 preterm; dHCP) | **Method:** fMRI (RS) | **Statistical Approach:** Stability-driven ROI-constrained connectome-based predictive modeling (CPM); dHCP; CBCL; ECBQ **Key Findings:** Prefrontal and temporoparietal systems were repeatedly implicated as predictive features. Neonatal fMRI established as a prognostic tool for socioemotional development.

Paper 7 Title: Neonatal sensory networks at birth predict cognitive, language, and motor outcomes at 18 months

 **fMRI (resting-state, CPM) · Neonates → 18 months**
· **Sensory networks · Bayley outcomes · dHCP**  **Read if you study:** neonatal brain predictors of cognitive or

language outcomes, or sensory network development in early life 💡 **In one sentence:** Neonatal visual and auditory cortex resting-state connectivity predicted Bayley cognitive, language, and motor scores at 18 months — establishing early sensory functional organization as a generalizable neurodevelopmental biomarker.

Authors: Mi Zou, Arun L. W. Bokde **Journal:** *bioRxiv* | **Date:** 2026-04-04 | **Preprint:** Yes (bioRxiv) **DOI:** <https://doi.org/10.64898/2026.04.04.716445> **First Author Institution:** Trinity College Dublin **Age Range:** Neonates → 18 months (N=402; dHCP) | **Method:** fMRI (RS) | **Statistical Approach:** ROI-degree-guided CPM; permutation testing; Bayley-III subscale prediction **Key Findings:** Sensory hub regions showed most generalizable predictive signal for later outcomes. Findings validated in held-out data across cognitive, language, and motor Bayley subscales.

Paper 4 ⭐ *Top Pick* **Title:** Explicitly nonlinear fMRI networks reveal hidden trajectories of infant brain development

🔬 **fMRI (resting-state, nonlinear ICA) · 0–12 months · Nonlinear connectivity · Default mode · Language networks** 📌 **Read if you study:** infant fMRI connectivity methods, or default mode / language network development in early life 💡 **In one sentence:** Nonlinear ICA revealed developmentally structured default mode, executive, language, and salience networks in infant fMRI that are completely invisible to standard linear methods.

Authors: Spencer Kinsey, Geethanjali Nagaboina, Prerana Bajracharya, Masoud Seraji, Zening Fu, Vince D. Calhoun, Sarah Shultz, Armin Iraji **Journal:** *bioRxiv* | **Date:** 2026-04-07 | **Preprint:** Yes (bioRxiv) **DOI:** <https://doi.org/10.64898/2026.04.07.716703> **First Author Institution:** Georgia State University / Emory University / TReNDS Center **Age Range:** 0–12 months | **Method:** fMRI (RS) | **Statistical Approach:** Data-driven nonlinear ICA; intrinsic connectivity networks;

developmental regression; comparison with linear ICA **Key**

Findings: First comprehensive developmental investigation of whole-brain nonlinear fMRI networks in human infants.

Neurobiologically structured nonlinear patterns present at birth; the nonlinear approach unveils insights into sensorimotor, default mode, executive, language, and saliency networks.

Paper 10 Title: Functional connectome harmonics capture early brain organization and maturity in neonates

 **fMRI (resting-state, FCH) · Neonates · Connectome harmonics · Brain maturity · Preterm · dHCP**  **Read**

if you study: neonatal brain maturity biomarkers, preterm neurodevelopment, or novel fMRI decomposition methods

 **In one sentence:** Functional connectome harmonic entropy, power, and energy predicted ~30% of neonatal postmenstrual age variance and distinguished term from preterm brains — offering new biomarkers of brain maturity at birth.

Authors: [Unverified — dHCP consortium] **Journal:** *bioRxiv* | **Date:** 2026-03-25 | **Preprint:** Yes (bioRxiv) **DOI:** <https://doi.org/10.64898/2026.03.25.714150> **Age Range:** Neonates (N=714; dHCP) | **Method:** fMRI (RS) | **Statistical Approach:** Functional connectome harmonics (FCH); entropy/power/energy metrics; age prediction; term vs. preterm comparison **Key Findings:** Adult-like sensory-to-multimodal gradient patterns present at birth. FCH energy and power higher in term-born vs. preterm neonates; entropy elevated in preterms.

Paper 32 Title: Simultaneous EEG-fMRI investigation of sound sequence processing in human neonates

 **Simultaneous EEG + fMRI · Neonates · Auditory processing · Mismatch response · Multimodal**  **Read**

if you study: neonatal auditory processing, EEG-fMRI multimodal methods, or early precursors of language  **In**

one sentence: Sleeping neonates engage distributed cortical and subcortical networks (including hippocampus and thalamus) for sound sequence deviance detection — revealing a multi-hub auditory network within days of birth.

Authors: [Unverified — full author list at DOI] **Journal:** *bioRxiv* | **Date:** 2026-01-09 | **Preprint:** Yes (bioRxiv) **DOI:** <https://doi.org/10.64898/2026.01.09.698718> **Age Range:** Neonates (N=25 full-term, 22 analyzable) | **Method:** Simultaneous EEG + fMRI | **Statistical Approach:** EEG mismatch response; fMRI BOLD contrasts; auditory oddball paradigm **Key Findings:** EEG mismatch response identified for deviant sounds but not omissions (local, not global, sequence regularity). fMRI reveals superior/ventromedial temporal lobe, precentral gyrus, thalamus, and parahippocampal cortex activated by auditory deviance in sleeping neonates.

4.4 fNIRS

Paper 1 🌟 *Top Pick — Priority Author* **Title:** Caregiver-infant interactions selectively shape emerging functional connectivity in the neonatal brain

 **fNIRS (naturalistic, home-based) · 1 month · Caregiver-infant · Social brain · Affectionate touch · Connectivity** 📌 **Read if you study:** early social brain development, caregiver-infant dynamics, fNIRS in naturalistic settings, or early adversity 💡 **In one sentence:** Affectionate caregiver touch and dyadic engagement at 1 month predicted stronger infant social brain network functional connectivity — demonstrating that caregiving quality shapes emerging connectivity in the first weeks of life.

Authors: L. Carnevali, B. Blanco, M. Rozhko, M. Weatherhead, M. H. Johnson, S. Lloyd-Fox **Journal:** *bioRxiv* | **Date:** 2026-05-13 | **Preprint:** Yes (bioRxiv) **DOI:** <https://doi.org/>

10.64898/2026.05.13.724776 **First Author Institution:** Birkbeck, University of London **Age Range:** 1 month | **Method:** fNIRS | **Statistical Approach:** Functional connectivity analysis; linear mixed models; dyadic interaction coding **Key Findings:** Repeated home-based fNIRS combined with coded caregiver-infant observations. Affectionate interactions associated with stronger connectivity, particularly within social brain networks, establishing early environmental shaping of neonatal functional architecture.

Paper 27 ★ *Priority Author* **Title:** Cross-paradigm fNIRS brain activity in 1-month-old infants across The Gambia and the United Kingdom

 **fNIRS (cross-cultural, multi-paradigm) · 1 month · Cross-cultural · Social stimuli · Global neuroscience · fNIRS** 📌 **Read if you study:** cross-cultural infant neuroscience, early social brain development, fNIRS methodology, or global health neuroscience 💡 **In one sentence:** One-month-old infants across The Gambia and the UK show both universal patterns and cross-cultural differences in fNIRS neural responses — establishing that environment shapes early brain function from the first month of life.

Authors: Isobel Greenhalgh, Borja Blanco, Chiara Bulgarelli, Ebrima Mbye, Ebou Touray, Maria Rozhko, Laura Katus, Nathan Hayes, Samantha McCann, Sophie E. Moore, Clare E. Elwell, Anna Blasi, Sarah Lloyd-Fox **Journal:** *Neurophotonics* | **Date:** 2026-01-20 | **Preprint:** No **DOI:** <https://doi.org/10.1117/1.NPh.13.S1.S13007> **First Author Institution:** University of Cambridge **Age Range:** 1 month | **Method:** fNIRS | **Statistical Approach:** fNIRS channel-level analysis; cross-cultural comparison; linear mixed models **Key Findings:** Cross-paradigm (social selectivity, habituation/novelty, functional connectivity) fNIRS study across markedly different environments reveals both universal early neural responses and cross-cultural

variation, informing how to design globally generalizable infant neuroscience research. **Key Figure:** fNIRS activation maps across paradigms for Gambian vs. UK infant cohorts.

Paper 29 Title: Singing to the newborn brain uncovers early traces of specialized neural networks

 **fNIRS · Neonates · Singing · Speech · Neural specialization · Communication** 📌 **Read if you study:** early auditory specialization, language and music in infancy, or fNIRS in newborns 💡 **In one sentence:** Both speech and song (vs. humming) elicit stronger right temporoparietal fNIRS responses in newborns — but no difference between speech and song — consistent with song's role in supporting language acquisition through rhythmic regularization.

Authors: Marino C., Gemignani J., Varnet L., and co-authors
Journal: *Communications Psychology* | **Date:** 2026-01-15 |
Preprint: No **DOI:** <https://doi.org/10.1038/s44271-026-00451-6>
Age Range: Neonates (N=40) | **Method:** fNIRS | **Statistical Approach:** Hemodynamic response analysis; within-subject contrasts (speech vs. song vs. humming) **Key Findings:** Early traces of specialized neural networks for communication present at birth. No neural distinction between speech and song, consistent with shared prosodic/rhythmic processing pathways.

Paper 11 Title: Spectral Geometry of Infant Resting-State fNIRS Connectivity: Bilingual vs Monolingual

 **fNIRS (resting-state, spectral geometry) · 0–12 months · Bilingualism · Resting-state · Connectivity · Manifold methods** 📌 **Read if you study:** bilingual language environments, resting-state fNIRS, or computational geometry approaches to connectivity 💡 **In one sentence:** Spectral geometry on resting-state fNIRS connectivity matrices discriminates bilingual from

monolingual infant brains (ROC-AUC=0.871), suggesting bilingual exposure induces detectable differences in infant brain network organization.

Authors: [Unverified — full author list at DOI] **Journal:** *bioRxiv* | **Date:** 2026-03-20 | **Preprint:** Yes (bioRxiv) **DOI:** <https://doi.org/10.64898/2026.03.20.707714> **Age Range:** 0–12 months (N=94) | **Method:** fNIRS | **Statistical Approach:** Spectral geometry on SPD correlation matrices; Grassmann manifold features; leave-one-out validation **Key Findings:** Shrinkage-regularized SPD correlation operators and spectral geometry methods applied to fNIRS achieve ROC-AUC of 0.811–0.871 for bilingual vs. monolingual classification, establishing a new computational approach for infant brain network analysis.

4.5 Language, Cognition & Developmental Trajectories

Paper 9 Title: Cascading periods of language-related brain plasticity across early childhood

 **fMRI (resting-state, Hurst exponent) · 10 months–15 years · Language sensitive periods · Cortical plasticity · BCP + HCP-D** 📌 **Read if you study:** language network development, cortical sensitive periods, or longitudinal fMRI methods across childhood 💡 **In one sentence:** Language-related cortical regions mature in a posterior-to-anterior cascade from 10 months to 15 years, and stronger language skills paradoxically associate with slower Hurst increases in early childhood — consistent with protracted plasticity as a developmental advantage.

Authors: Monica E. Ellwood-Lowe, Monami Nishio, Alexander J. Dufford, Michael Arcaro, Theodore D. Satterthwaite, Allyson P. Mackey **Journal:** *bioRxiv* | **Date:** 2026-03-27 | **Preprint:** Yes (bioRxiv) **DOI:** <https://doi.org/10.64898/2026.03.27.714739> **First Author Institution:** University of Pennsylvania **Age Range:** 10 months–15 years (BCP n=222, 458 obs; HCP-D n=324) |

Method: fMRI | **Statistical Approach:** Hurst exponent analysis; growth curve modeling; cross-dataset validation **Key Findings:** Thalamic Hurst plateaus in the first year; cortical Hurst plateaus around age 9. Confirmed across two independent large datasets spanning birth to adolescence.

Paper 15 Title: The emergence of the language system in the toddler brain

 **fMRI (awake, task-based) · 19–36 months · Language networks · Left lateralization · Toddlers** 
Read if you study: toddler language development, left-hemisphere lateralization, or awake toddler fMRI  **In one sentence:** Left frontal and temporal cortex in toddlers aged 19–36 months already responds more to comprehensible than reversed language — demonstrating adult-like lateralization years earlier than previously shown with fMRI.

Authors: [Unverified — full author list at DOI] **Journal:** *bioRxiv* | **Date:** 2026-03-02 | **Preprint:** Yes (bioRxiv) **DOI:** <https://doi.org/10.64898/2026.02.26.707550> **Age Range:** 19–36 months (N=29; Sesame Street audio paradigm) | **Method:** fMRI (awake, task-based) | **Statistical Approach:** Univariate fMRI contrasts; left vs. right laterality analysis **Key Findings:** Brain organization for language already adult-like in its left-hemispheric lateralization by 19–36 months, though response magnitude weaker than in adults. Foundational left-lateralized language networks established during the toddler period.

Paper 8 Title: Developmental links between play behavior and brain network integration across infancy to middle childhood

 **fMRI (resting-state) · 10 months–12 years · Play behavior · DMN · Cognitive control · Salience network**  **Read if you study:** play and cognitive development, default mode network maturation, or brain-behavior

relationships in childhood 💡 **In one sentence:** Across three studies spanning 10 months to 12 years, more frequent play consistently associated with greater integration between default mode, salience, and cognitive control networks — a finding robust to Montessori vs. traditional schooling.

Authors: [Multiple institutions — bioRxiv] **Journal:** *bioRxiv* | **Date:** 2026-03-28 | **Preprint:** Yes (bioRxiv) **DOI:** <https://doi.org/10.64898/2026.03.26.714609> **Age Range:** 10 months–12 years | **Method:** fMRI (RS) | **Statistical Approach:** Resting-state network analysis; between-network coupling; three independent studies (N=143, 108, 45) **Key Findings:** Play in infancy associated with stronger DMN–cognitive control network coupling; findings replicate across samples and schooling contexts.

Paper 6 Title: Separable Brain Maturation Patterns Mediate Exposome Influences on Cognitive Development: A Longitudinal Study

🔬 **Multimodal (4,882 brain features) · 9–15 years · Exposome · Cognitive development · Mediation · Multivariate** 📌 **Read if you study:** environmental influences on cognitive development, brain-environment mediation, or multivariate developmental neuroimaging 💡 **In one sentence:** Factor analysis of domain-level brain mediation reveals four latent neural patterns each linked to distinct exposome domains, with the family environment domain showing the strongest correspondence with brain maturation and cognitive ability.

Authors: Jiadong Yan, Judy Chen, Bin Wan, Paule Joanne Toussaint, Alan C. Evans, Sherif Karama **Journal:** *bioRxiv* | **Date:** 2026-04-05 | **Preprint:** Yes (bioRxiv) **DOI:** <https://doi.org/10.64898/2026.04.04.716497> **First Author Institution:** McGill University **Age Range:** 9–15 years (N=1,112 longitudinal) | **Method:** Mixed (4,882 multimodal brain features; 112 exposome

variables) | **Statistical Approach:** Mediation analysis; factor analysis; 6 exposome domains **Key Findings:** Different exposome domains influence cognitive development through partially separable brain mechanisms, suggesting targeted environmental interventions would have domain-specific neural effects.

5. Methods Overview

Method	# Papers	% of Total
EEG (resting-state, oscillatory, or VEP)	14	41%
fMRI (resting-state)	11	32%
fMRI (task-based / awake)	3	9%
fNIRS	4	12%
Simultaneous EEG + fMRI	1	3%
MEG	1	3%

6. Statistical Approaches Trending This Period

- FOOOF / specparam aperiodic decomposition** — Dominant shift across EEG and fMRI. Multiple groups apply aperiodic decomposition to infant/toddler EEG (papers 12, 24, 31, 34) and now also to BOLD fMRI signals (paper 25), linking the exponent to autism, nutrition, adversity, and normative development.
- Connectome-based predictive modeling (CPM)** — Applied to neonatal fMRI for prospective prediction of 18-month outcomes (papers 3, 7). Stability-driven ROI-constrained CPM is establishing itself as the standard for early-life prognostic connectivity research.
- Nonlinear and higher-order connectivity** — Kinsey et al. (nonlinear ICA; paper 4) and Li et al. (tri-FNC tensor decomposition; paper 28) demonstrate that standard linear correlation methods miss major classes of developmentally structured fMRI connectivity.

4. **Network-based graph-theoretic EEG analysis** — Chung et al. (paper 30) apply partial correlation networks and graph-theoretic metrics to longitudinal EEG, demonstrating network topology as a sensitive measure of adversity effects.
5. **Hurst exponent / long-range temporal correlations** — Ellwood-Lowe et al. (paper 9) apply Hurst exponent as an in vivo plasticity proxy to map language sensitive period cascades across two large longitudinal datasets.
6. **Spectral geometry on connectivity manifolds** — Paper 11 applies SPD correlation operators and Grassmann manifold features to resting-state fNIRS, achieving AUC=0.871 for bilingual/monolingual classification — a novel computational geometry approach.
7. **Burst-resolved oscillatory analysis** — Vanneau et al. (paper 5) demonstrate that decomposing EEG into burst amplitude vs. temporal stability reveals mechanistic information about autism-related alpha dysregulation lost in standard spectral power analysis.
8. **Functional connectome harmonics (FCH)** — Paper 10 introduces FCH-derived entropy, power, and energy metrics as novel neonatal brain maturity biomarkers explaining ~30% of postmenstrual age variance.

7. Notable Authors & Labs This Period

Researcher	Institution	Specialty	Paper(s)
Charles A. Nelson	BCH / Harvard	EEG, early adversity	30, 31, 34
Carol L. Wilkinson	BCH / Harvard	EEG, autism biomarkers	30, 31, 34
Helen Tager-Flusberg	Boston University	Autism, language	30, 31, 34
Sarah Lloyd-Fox	Univ. of Cambridge / Birkbeck	fNIRS, caregiver-infant	1, 27
M. H. Johnson	Birkbeck, Univ. of London	Infant cognition, fNIRS	1
Rhodri Cusack		Awake infant fMRI	23

Researcher	Institution	Specialty	Paper(s)
	Trinity College Dublin		
Arun L. W. Bokde	Trinity College Dublin	Neonatal fMRI, connectomics	3, 7
Vince D. Calhoun / Armin Iraj	TReNDS / Georgia State	Nonlinear connectivity, ICA	4, 28
Duncan E. Astle	Univ. of Cambridge	Network neuroscience, dev	20
Allyson P. Mackey	Univ. of Pennsylvania	Plasticity, language	9
Nathan A. Fox	Univ. of Maryland	EEG, early development	33
Terje Falck-Ytter	Karolinska Institutet	Infant EEG, autism	17
John J. Foxe / Sophie Molholm	Univ. of Rochester	HD-EEG, autism	5
J. Christopher Edgar	CHOP	Pediatric MEG/EEG	12

8. Preprints to Watch

1. **Paper 1 — Caregiver-infant interactions shape fNIRS connectivity at 1 month** (bioRxiv, May 2026) — Lloyd-Fox / M. H. Johnson group; naturalistic home-based fNIRS at 1 month. URL: <https://doi.org/10.64898/2026.05.13.724776>
2. **Paper 3 — Neonatal RS-FC Predicts Socioemotional Outcomes at 18 Mo** (bioRxiv, April 2026) — CPM in dHCP neonates; watch for peer-reviewed submission. URL: <https://doi.org/10.64898/2026.04.21.719787>
3. **Paper 4 — Explicitly Nonlinear fMRI Networks** (bioRxiv, April 2026) — Methodological landmark; likely to generate significant follow-up. URL: <https://doi.org/10.64898/2026.04.07.716703>
4. **Paper 9 — Cascading Language Plasticity** (bioRxiv, March 2026) — BCP + HCP-D; Hurst exponent for sensitive period mapping. URL: <https://doi.org/10.64898/2026.03.27.714739>

5. Paper 14 — Visual Cortical Response Variability in Autism-Risk Infants (bioRxiv, March 2026) — IBIS Network; P1 latency variability predicts 24-month outcomes. URL: <https://doi.org/10.64898/2026.03.05.709374>

9. Complete Paper List

#	Citation	DOI	Date
1	Carnevali L. et al. (Lloyd-Fox, Johnson). <i>Caregiver-infant interactions shape neonatal fNIRS connectivity</i> . bioRxiv.	10.64898/2026.05.13.724776	2026-05-13
2	Khandelwal P. et al. <i>Cortical reconstruction of postmortem ex vivo infant MRI</i> . bioRxiv.	10.64898/2026.05.07.722301	2026-05-09
3	Zou M., Bokde A.L.W. <i>Neonatal RS-FC Predicts Socioemotional Outcomes at 18 Mo</i> . bioRxiv.	10.64898/2026.04.21.719787	2026-04-21
4	Kinsey S. et al., Iraj A. <i>Explicitly nonlinear fMRI networks reveal hidden trajectories</i> . bioRxiv.	10.64898/2026.04.07.716703	2026-04-07
5	Vanneau T. et al. <i>Instability of Alpha Oscillatory States in Autism</i> . bioRxiv.	10.64898/2026.04.03.716324	2026-04-07
6	Yan J. et al. <i>Separable Brain Maturation Patterns Mediate Exposome Influences on Cognition</i> . bioRxiv.	10.64898/2026.04.04.716497	2026-04-05
7	Zou M., Bokde A.L.W. <i>Neonatal sensory networks predict cognitive/language/motor at 18 Mo</i> . bioRxiv.	10.64898/2026.04.04.716445	2026-04-04
8	[Authors]. <i>Developmental links between play behavior and brain network integration</i> . bioRxiv.	10.64898/2026.03.26.714609	2026-03-28
9	Ellwood-Lowe M.E. et al. <i>Cascading periods of language brain plasticity</i> . bioRxiv.	10.64898/2026.03.27.714739	2026-03-27
10	[dHCP consortium]. <i>Functional connectome</i>	10.64898/2026.03.25.714150	2026-03-25

#	Citation	DOI	Date
	<i>harmonics in neonates.</i> bioRxiv.		
11	[Authors]. <i>Spectral Geometry of Infant RS-fNIRS: Bilingual vs. Monolingual.</i> bioRxiv.	10.64898/2026.03.20.707714	2026-03-20
12	Green H.L. et al. <i>Resting-state periodic and aperiodic oscillations birth–preschool.</i> Dev. Cogn. Neurosci.	10.1016/j.dcn.2026.101709	2026-03-14
13	Leroy F. et al. <i>Dynamics of the attentional blink in preverbal infants.</i> PNAS.	10.1073/pnas.2526752123	2026-03-10
14	Dickinson A. et al. (IBIS). <i>Visual Cortical Response Variability at High Familial Likelihood for Autism.</i> bioRxiv.	10.64898/2026.03.05.709374	2026-03-09
15	[Authors]. <i>Emergence of the language system in the toddler brain.</i> bioRxiv.	10.64898/2026.02.26.707550	2026-03-02
16	Beaugrand M. et al. <i>Tracing infant sleep neurophysiology longitudinally 3–6 months.</i> npj Bio Timing Sleep.	10.1038/s44323-026-00071-7	2026-03-01
17	Hardiansyah I. et al. <i>Visual cortex FC links to motion/autism at 5 months.</i> Sci. Reports.	10.1038/s41598-026-42048-3	2026-02-28
18	[Authors]. <i>Non-linear development of auditory neural responses in the first year.</i> bioRxiv.	10.64898/2026.02.20.706969	2026-02-20
19	Sa de Almeida J. et al. <i>Regional BOLD variability and microstructural maturation in preterm.</i> Nature Comms.	10.1038/s41467-026-71415-x	2026-02-15
20	Epihova G. et al. (Astile). <i>Fronto-Temporal Cortex subcortical connectivity in utero.</i> J. Neuroscience.	10.1523/JNEUROSCI.0832-25.2025	2026-02-12
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